

GARDELEGEN, Germany

WMO: 103590

Lat: 52.5128N

Lon: 11.3942E

Elev: 157

StdP: 14.61

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.6	12.5	-0.5	5.4	7.3	5.8	7.5	14.7	29.4	43.2	25.7	42.5	1.4	30	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	85.6	66.2	82.0	64.5	78.7	63.2	67.7	81.0	65.9	78.6	64.3	76.3	7.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.0	86.6	72.6	60.9	80.3	71.7	59.1	75.3	69.9	32.1	81.2	30.7	78.8	29.4	76.3	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	20.0	17.3	DB	1.5	91.4	12.6	4.2	-7.5	94.4	-14.8	96.9	-21.9	99.2	-31.0	102.3	(4)
(5)			WB	0.5	71.6	12.2	1.8	-8.3	72.9	-15.5	73.9	-22.3	74.9	-31.2	76.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	32.7	31.7	39.7	45.2	54.3	59.7	63.6	63.1	56.4	48.7	40.2	35.8	(6)
(7)		DBStd	13.27	10.74	9.85	7.02	6.60	6.60	6.22	5.76	5.52	4.82	6.67	6.68	7.47	(7)
(8)		HDD50	2425	536	512	324	173	32	2	0	0	4	102	298	442	(8)
(9)		HDD65	6450	1001	931	785	594	335	184	99	104	262	505	744	906	(9)
(10)		CDD50	1585	1	1	3	29	166	292	421	407	196	63	5	1	(10)
(11)		CDD65	133	0	0	0	0	4	24	55	46	3	1	0	0	(11)
(12)		CDH74	1954	0	0	0	1	125	401	753	611	56	7	0	0	(12)
(13)		CDH80	526	0	0	0	0	8	106	219	187	6	0	0	0	(13)
(14)	Wind	WSAvg	6.9	8.7	7.3	7.8	6.5	5.9	6.0	6.0	5.6	6.7	6.5	7.5	8.1	(14)
(15)	Precipitation	PrecAvg	22.9	1.9	1.4	1.5	1.3	2.0	2.2	2.8	2.4	2.0	1.8	1.8	1.9	(15)
(16)		PrecMax	31.7	4.3	3.0	3.7	2.8	5.4	3.8	5.7	5.2	5.2	5.4	3.9	4.1	(16)
(17)		PrecMin	16.2	0.0	0.4	0.2	0.1	0.7	0.9	0.8	0.7	0.3	0.4	0.0	0.7	(17)
(18)		PrecStd	3.9	0.9	0.7	0.9	0.7	1.3	0.9	1.4	1.2	1.1	1.1	0.9	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.1	58.6	64.1	71.9	80.9	88.3	88.8	89.8	80.4	73.0	58.2	54.8	(19)
(20)			MCWB	49.7	50.9	51.8	56.9	61.1	65.6	67.4	68.3	63.2	63.0	51.8	49.7	(20)
(21)		2%	DB	49.6	52.3	57.6	66.9	77.7	82.7	85.7	85.2	74.4	66.4	54.4	51.8	(21)
(22)			MCWB	46.2	46.9	49.4	54.2	59.5	64.2	66.7	65.9	61.4	57.8	50.6	47.9	(22)
(23)		5%	DB	47.5	47.4	53.2	62.4	74.0	78.7	82.3	80.6	69.8	62.2	52.4	49.4	(23)
(24)			MCWB	44.2	43.4	46.3	51.9	58.2	62.2	65.2	64.4	59.0	55.5	49.0	46.3	(24)
(25)		10%	DB	45.0	43.9	50.1	58.3	70.0	74.5	78.5	76.5	66.3	59.2	50.2	46.8	(25)
(26)			MCWB	41.9	40.2	45.0	49.4	56.6	60.4	64.0	63.2	57.8	53.6	47.3	44.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.9	51.2	52.9	58.2	63.9	69.1	70.5	70.1	65.2	63.1	52.9	50.1	(27)
(28)			MCD B	53.2	57.3	61.1	69.8	76.2	79.5	85.3	86.0	76.6	71.3	55.6	53.7	(28)
(29)		2%	WB	46.9	46.8	50.0	55.1	61.3	65.4	68.1	67.3	62.6	59.1	51.2	48.3	(29)
(30)			MCD B	49.4	51.7	56.3	64.7	74.2	79.9	81.5	80.7	72.2	66.0	54.1	51.2	(30)
(31)		5%	WB	44.4	43.5	47.6	52.6	59.3	63.3	66.2	65.7	60.5	56.1	49.4	46.5	(31)
(32)			MCD B	47.3	47.4	52.2	61.1	70.8	76.2	79.2	77.4	67.9	61.3	52.3	49.1	(32)
(33)		10%	WB	42.1	40.5	45.2	50.2	57.1	61.2	64.3	64.0	58.6	54.1	47.3	44.1	(33)
(34)			MCD B	44.9	43.5	49.7	57.4	68.6	73.0	76.8	74.7	65.2	58.7	50.1	46.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.8	13.1	14.0	18.7	21.4	19.7	20.8	20.7	17.0	15.2	10.5	8.7	(35)
(36)			MCD BR	10.3	16.8	21.2	26.5	29.3	29.0	29.5	29.5	24.8	21.1	13.7	11.1	(36)
(37)			MCWBR	8.7	13.0	14.7	15.8	15.3	14.2	14.6	14.7	14.9	14.4	11.2	9.1	(37)
(38)		5% WB	MCD BR	9.4	16.1	19.6	23.8	26.2	25.4	26.3	24.7	20.5	19.5	13.0	10.5	(38)
(39)	MCWBR		8.3	12.8	14.2	14.7	14.6	13.5	13.9	13.3	13.0	14.1	10.9	9.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.317	0.340	0.376	0.413	0.413	0.410	0.429	0.422	0.386	0.360	0.337	0.311	(40)	
(41)		taud	2.354	2.339	2.254	2.205	2.251	2.279	2.239	2.272	2.355	2.406	2.394	2.352	(41)	
(42)		Ebn at Noon	213	243	255	259	266	268	259	254	250	232	203	192	(42)	
(43)		Edh at Noon	19	26	34	41	41	40	41	38	31	24	19	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	207	430	770	1266	1559	1682	1578	1336	982	555	249	159	(44)	
(45)		RadStd	21	60	73	149	179	133	219	109	112	57	27	12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)	+0.68	N/A	N/A	N/A	N/A	+0.95	N/A	-284	N/A	N/A

Nomenclature: See separate page